Hamed Barzamini

Location: Chicago Metropolitan Area
Phone: 815 901 8364
Email: h.barzamini@niu.edu
LinkedIn: linkedin.com/in/hamed-b-a0943280/

Education

2020–Present **Ph.D. in Computer Science**, Northern Illinois University, GPA: 3.83/4.
Dissertation: (In Progress) *V-REX: Unified Semantics-Driven Requirements Engineering Framework,* Advisor: Dr. Mona Rahimi

2012–2015 **Master of Science in Computer Science**, Shahid Beheshti University, GPA: 3.45/4.
Thesis: *AMI Theft Detection: A Data-Driven Energy Loss View*

2008–2012 **Bachelor of Science in Computer Science**, University of Mazandaran, GPA: 3.81/4, Graduated with the highest GPA.

Publications

Journals

Shahzad, M., **Barzamini, H.**, Wilson, J., Alhoori, H., & Rahimi, M. (2025). Dynamic domain analysis for predicting concept drift in engineering AI-enabled software. *Journal of Data and Information Science*, 10(2), 124–151.

Barzamini, H., Shahzad, M., Alhoori, H., & Rahimi, M. (2022). A multi-level semantic web for hard-to-specify domain concept, Pedestrian, in ML-based software. Requirements Engineering, 1-22.

Barzamini, H., & Ghassemian, M. (2019). Comparison analysis of electricity theft detection methods for advanced metering infrastructure in smart grid. International Journal of Electronic Security and Digital Forensics, 11(3), 265-280.

Conferences

Barzamini, H., Ramesh, S., Adiththan, A., Peranandam, P., & Rahimi, M. (2025). Specifying Operational Design Domain in Autonomous Driving for Comprehensive Data Evaluation. In 2025 IEEE 33rd International Requirements Engineering Conference (RE), 43–55.

Barzamini, H., Brockmann, A., Nazaritiji, F., Ferdowsi, H., & Rahimi, M. (2025). An AI-driven Requirements Engineering Framework Tailored for Evaluating AI-Based Software. Accepted at CAIN 2025 — 4th International Conference on AI Engineering – Software Engineering for AI.

Barzamini, H., & Rahimi, M. (2022, October). B-AIS: An Automated Process for Black-box Evaluation of Visual Perception in AI-enabled Software against Domain Semantics. In Proceedings of the 37th IEEE/ACM International Conference on Automated Software Engineering (pp. 1–13).

Barzamini, H., & Rahimi, M. (2022, August). CADE: The Missing Benchmark in Evaluating Dataset Requirements of AI-enabled Software. In 2022 IEEE 30th International Requirements Engineering Conference (RE) (pp. 64–76). IEEE.

Barzamini, H., Rahimi, M., Shahzad, M., & Alhoori, H. (2022, May). Improving Generalizability of ML-enabled Software Through Domain Specification. In Proceedings of the 1st International Conference on AI Engineering: Software Engineering for AI (pp. 181–192).

Professional Experience

May 2021 – Present **Research Assistant**, Northern Illinois University, DeKalb, US.
Advisor: Dr. Mona Rahimi

- **Intel oneAPI Student Ambassador:** Optimized AI workflows using oneDNN and oneDAL, reducing processing time by **40%**.

- **Optimized AI Requirement Analytics:** Engineered a multi-level semantic web to identify domain gaps, revealing **80%** dataset deficiencies in pedestrian variations. Enhanced dataset completeness, leading to a **4.9%** boost in model accuracy.
- **Enhanced AI Model Generalizability:** Leveraged LLM-driven semantic specifications and diffusion-based generative AI to refine training data, reducing model bias by **7.3%** and improving dataset completeness, leading to a **4.9%** increase in model accuracy.
- **Developed a Semantic Benchmark for Dataset Evaluation:** Engineered a systematic dataset evaluation process using domain-driven embedding techniques, identifying **25%** deficiencies in dataset alignment with real-world specifications.
- **Refined AI Perception with Evaluation Benchmark:** Leveraged XAI techniques, black-box evaluation, and a systematic benchmarking process to identify weaknesses in model performance, enhancing AI perception of domain concepts. Achieved an **F2-score of 95%** for pedestrians and **72%** for aircraft.
- **Designed Concept Drift Detection Mechanisms:** Built domain shift detection models using adaptive feature mapping, improving AI robustness against changing conditions.
- **Collaborated on Research Proposals:** Contributed to three research proposals (NSF and General Motors) focusing on AI-enabled perception systems, requirements engineering frameworks, and safety-critical software systems.

Jan 2024 – May 2024 **Research and Development Internship**, General Motors (GM), Warren, MI, USA.

- **Developed an AI-driven framework:** Designed a method leveraging prompt learning and RAG-based knowledge graphs to specify operational design domains (ODD) in autonomous driving, enhancing dataset validation and system robustness.
- **Evaluated dataset completeness:** Applied the framework to assess autonomous driving datasets, improving alignment with ODD specifications by **35%**, ensuring higher fidelity in safety-critical scenarios.
- **Presented Research to GM R&D Team:** Provided insights on dataset validation, shaping discussions on ADS safety standards.

Nov 2015 – Apr 2020 **Additional Research and Development Projects**

- **Developed AI-Driven Customer Analytics:** Built an AI-powered recommendation system, increasing customer engagement in malls by **20%**.
- **Optimized Web Experience with Big Data:** Improved content personalization algorithms, enhancing user satisfaction by **25%**.
- **Accelerated Pedestrian Detection Algorithms:** Applied parallelization techniques, boosting detection speed by **35%**.
- **Designed Adaptive User Profiling Framework:** Increased behavioral prediction accuracy by **30%** using machine learning.
- **Engineered Real-Time Anomaly Detection:** Reduced false alerts in operational time-series metrics by **40%**.
- **Automated DevOps Pipelines with GitLab:** Decreased software deployment time by **50%**.
- **Built a Big Data Pattern Detection Module:** Improved data processing efficiency by **15%**.
- **Led AI-Driven Product Innovation:** Spearheaded AI-based solutions, accelerating software development cycles.
- **Integrated Software Engineering Best Practices:** Standardized workflows, improving code maintainability and team efficiency.
- **Developed Web Log Analysis Tool:** Reduced issue resolution time by **35%** through automated log parsing.
- **Applied Test-Driven Development (TDD):** Minimized software defects by **20%** with rigorous testing strategies.

Teaching Experience

Aug 2020 – May 2021 **Teaching Assistant**, Northern Illinois University, Department of Computer Sciences, DeKalb, US.

- *Computer Architecture and Systems Organization*: Review work, and Give Feedback to Enhance Concept Comprehension.
- *Programming Principles in C++*: Evaluate Assignments, Providing Constructive Feedback for Coding Skills and Collaborative Problem-Solving.

Aug 2013 – Dec 2013 **Teaching Assistant**, Shahid Beheshti University, Department of Computer Sciences and Engineering, Tehran, Iran.

- *Software Engineering*: Assist Students with Software Engineering Project Development, Review and Provide Feedback on Student Code Submissions.

Aug 2011 – Dec 2011 **Lab Tutor**, University of Mazandaran, Department of Computer Engineering, Babolsar, Iran.

- *Operating System*: Support Students with Practical OS Lab Exercises and, Guide Students in Debugging OS Lab Challenges.

Grants, Fellowships & Professional Service

Grants & Fellowships

Awarded — *Requirements Domain Specifications for Machine-Learned Software Components* (NSF CISE/CCF, EAGER; **Award #2124606**). PI: Dr. Mona Rahimi. **Research Assistant**. 2021–2024.
Submitted — *Automated Assessment, Enhancement & Explanation of AI Vision-based Perception in ADAS* (NSF with General Motors). PI: Dr. Mona Rahimi. **Research Assistant**. Status: Pending.
Submitted — *A Systematic Requirements Engineering Approach for Semantic- and Context-related Risk Management in AI-enabled Perception.* NSF. PI: Dr. Mona Rahimi. **Research Assistant**. Status: Pending.

Professional Service

Reviewer: IEEE Intl. Requirements Engineering Conf. (RE 2022).
Student Volunteer: 1st Intl. Conf. on AI Engineering: SE4AI (CAIN 2022); RE 2022.

Technical Skills

Python and C/C++; PyTorch and TensorFlow for deep learning (transformers/diffusion) with CUDA; NumPy, Pandas, and scikit-learn for analysis; vector search and retrieval (FAISS/HNSW) for IR/ML systems; SQL (PostgreSQL/MySQL); Git and Docker for CI-ready reproducibility; Linux development; OpenCV for vision tasks.

References

- **Dr. Mona Rahimi**
  Associate Professor
  Department of Computer Science
  Northern Illinois University
  `mrahimi1@niu.edu`
- **Dr. Hamed Alhoori**
  Associate Professor
  Department of Computer Science
  Northern Illinois University
  `alhoori@niu.edu`
- **Dr. Farzin Ferdowsi**
  Associate Professor
  Department of Electrical Engineering
  Northern Illinois University
  `hferdowsi@niu.edu`
- **Dr. Ali Cheraghian**
  Senior AI/ML Scientist
  Macquarie University
  Adjunct Faculty, Australian National University
  `Ali.cheraghian@mq.edu.au`